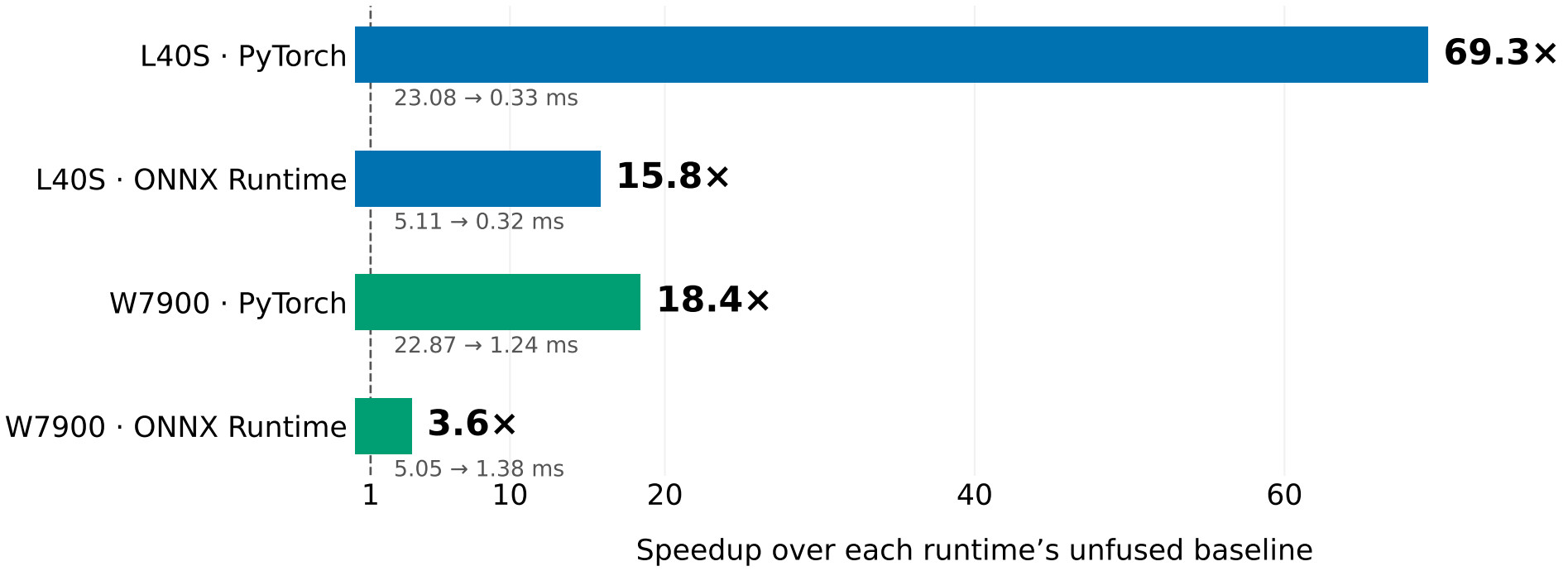
Fused attention: how much faster?

L40S and W7900 · FP16 · sequence length 2,560 · batch 4 · 16 heads · head dim 16



PyTorch: MATH → FLASH_ATTENTION backend.

ONNX Runtime / L40S: legacy SDPA decomposition → Attention-24 (FlashAttention).

ONNX Runtime / W7900: same legacy graph, default → explicit MIGraphX attention fusion.

Unmasked attention microbenchmark; median steady-state latency. ORT uses device I/O binding (no host transfers).

PyTorch 2.9.1 · ORT 1.29 CUDA / 1.23.2 MIGraphX · ROCm 7.2.4. Gains are within each stack, not end-to-end MLPF speedups.